

WANGARATTA, Australia

WMO: 948890

Lat: 36.4206S

Lon: 146.3056E

Elev: 503

StdP: 14.43

Time Zone: 10.00 (AUS)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.6	30.5	22.8	17.5	72.6	26.3	20.7	58.4	21.9	50.3	19.8	50.9	1.8	210	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.8	99.2	64.8	95.0	64.3	91.5	63.5	70.7	81.6	68.9	81.2	67.4	81.7	11.1	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	105.7	72.9	65.9	97.4	71.8	63.0	87.7	70.4	34.9	80.5	33.4	80.8	32.1	81.7	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	18.1	15.9	DB	25.6	106.2	2.2	3.7	24.0	108.8	22.8	111.0	21.5	113.0	19.9	115.7	(4)
(5)			WB	25.3	73.4	2.2	2.8	23.7	75.4	22.4	77.1	21.2	78.7	19.7	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	73.6	72.1	66.5	58.2	51.2	46.9	46.1	47.2	51.5	56.6	63.7	68.8	(6)	
	DBStd	11.08	6.69	6.23	6.03	5.17	4.64	4.04	3.72	3.75	4.34	5.38	6.38	6.53	(7)	
	HDD50	422	0	0	0	3	41	109	130	104	30	5	0	0	(8)	
	HDD65	3200	10	11	54	212	428	543	586	551	405	267	100	33	(9)	
	CDD50	3512	733	620	511	248	79	16	10	17	75	209	410	584	(10)	
	CDD65	814	278	211	100	7	0	0	0	0	6	60	152		(11)	
	CDH74	12646	4069	2872	1541	225	3	0	0	0	14	277	1179	2466	(12)	
	CDH80	6023	2232	1423	608	34	0	0	0	0	1	75	477	1173	(13)	
(14) Wind	WSAvg	6.2	7.7	6.8	6.2	5.0	4.7	4.7	5.4	5.8	6.4	6.5	7.2	7.9	(14)	
(15) Precipitation	PrecAvg	25.7	1.7	2.1	1.5	1.5	2.1	2.7	2.8	2.5	2.4	2.1	2.2	2.0	(15)	
	PrecMax	38.6	5.4	7.4	4.3	4.0	4.5	5.5	5.3	4.7	6.6	6.0	4.8	5.2	(16)	
	PrecMin	11.7	0.0	0.0	0.0	0.0	0.0	1.2	0.6	0.3	0.6	0.0	0.6	0.2	(17)	
	PrecStd	6.6	1.5	2.1	1.1	1.1	1.4	1.2	1.2	1.3	1.4	1.6	1.2	1.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	106.1	102.5	96.5	84.2	72.0	64.2	61.2	67.2	75.6	87.8	95.9	101.8	(19)	
		MCWB	65.5	66.0	64.2	61.3	57.3	54.2	52.6	56.2	59.0	60.4	62.8	63.5	(20)	
	2%	DB	100.6	96.7	91.0	79.2	68.2	60.8	58.8	62.7	70.2	80.9	89.9	95.4	(21)	
		MCWB	65.0	65.2	63.5	59.8	56.2	52.8	51.7	53.3	57.6	59.5	61.6	63.0	(22)	
	5%	DB	95.4	92.8	86.3	76.3	65.2	58.7	56.5	59.3	66.5	76.6	85.7	91.1	(23)	
		MCWB	64.3	64.6	62.3	59.0	55.3	51.8	50.4	51.2	55.5	59.1	61.4	61.8	(24)	
	10%	DB	91.5	89.0	82.7	72.9	62.3	56.0	54.0	57.0	63.7	71.7	80.8	86.3	(25)	
		MCWB	64.1	63.7	61.4	57.7	54.2	50.9	49.0	50.3	54.1	57.7	60.5	61.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	73.6	70.3	65.7	61.4	57.6	55.5	57.4	62.1	66.2	69.8	70.9	(27)	
		MCDB	81.5	83.2	79.2	75.5	64.9	59.6	57.9	63.2	69.5	76.8	81.3	76.7	(28)	
	2%	WB	70.3	70.8	67.6	62.9	58.8	55.0	52.9	54.8	59.5	63.4	67.0	68.1	(29)	
		MCDB	83.1	81.9	78.2	71.0	64.0	58.1	56.4	60.4	67.6	72.4	78.7	79.1	(30)	
	5%	WB	68.6	68.8	65.7	61.0	57.0	53.1	51.4	52.5	57.2	61.2	64.9	66.0	(31)	
		MCDB	83.4	81.4	79.5	71.1	62.7	56.7	54.9	58.2	64.5	70.6	76.4	79.5	(32)	
	10%	WB	66.9	67.0	64.0	59.3	55.3	51.5	50.1	50.8	55.1	59.3	63.0	64.2	(33)	
		MCDB	84.5	82.3	78.4	69.9	60.8	55.2	53.1	55.7	61.8	69.7	74.4	80.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	30.8	29.3	28.7	27.3	22.6	19.3	17.5	19.8	22.9	26.8	28.7	30.4	(35)	
		MCDBR	38.2	35.5	34.6	33.0	27.6	24.4	21.7	25.6	29.6	37.0	37.9	38.5	(36)	
		MCWBR	10.6	10.8	12.5	15.2	16.0	16.5	14.8	16.7	17.4	17.8	13.1	11.8	(37)	
	5% WB	MCDBR	27.2	25.4	26.7	26.2	20.6	18.8	17.0	21.9	25.2	27.7	26.5	27.6	(38)	
		MCWBR	10.0	9.8	11.4	13.2	13.1	13.7	12.3	14.9	15.9	15.7	11.9	11.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.356	0.349	0.322	0.315	0.302	0.294	0.290	0.298	0.331	0.331	0.346	0.349	(40)	
	taud		2.455	2.496	2.569	2.551	2.561	2.584	2.595	2.553	2.445	2.475	2.472	2.469	(41)	
	Ebn at Noon		310	305	302	285	271	265	273	287	293	308	312	314	(42)	
	Edh at Noon		37	35	30	27	23	21	22	26	33	35	37	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2433	2129	1760	1257	852	646	695	980	1409	1881	2191	2431	(44)	
	RadStd		139	130	90	67	65	83	47	84	141	166	116	108	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.81	N/A	N/A	+2.86	N/A	N/A	N/A	N/A	N/A	+196
(47) Regional (35 neighbors)		+0.81	N/A	N/A	+1.75	N/A	N/A	N/A	-175	+275	+126

Nomenclature: See separate page